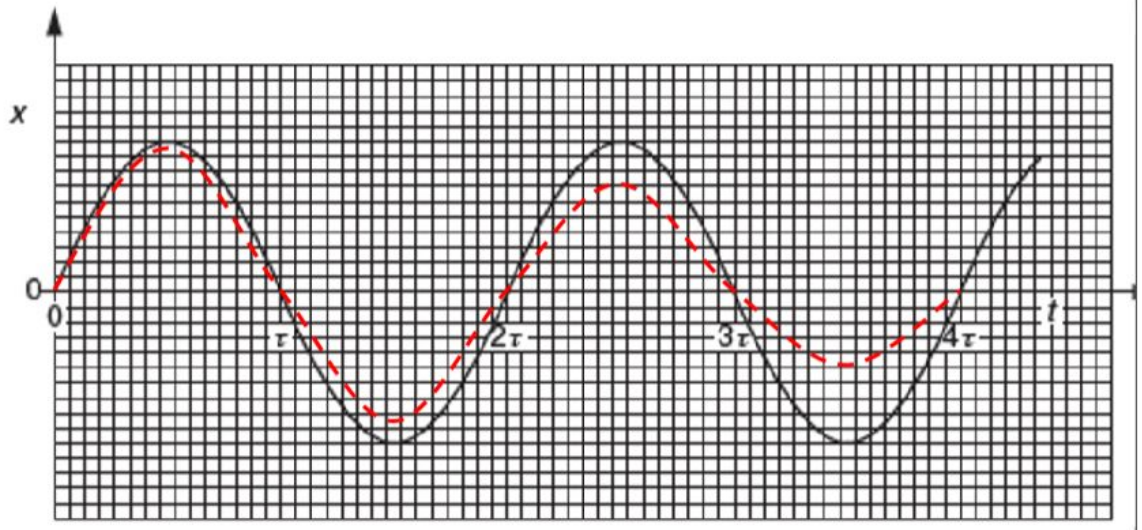
	Kinetic energy store is zero so speed is zero.	B1
4ai	g and r are constants, so a is proportional to x. The negative sign shows a and x are in opposite directions.	B1 B1
4aii	$\omega^2 = \frac{g}{r}$ and $\omega = \frac{2\pi}{T}$ $\omega^2 = \frac{9.81}{0.28} = 35$ $T = \frac{2\pi}{\sqrt{35}} = 1.06$ s Time interval = half a period = $T/2$ = 1.06/2 = 0.53 s	 C1 C1 A1
4b	 period is constant, displacement decreases successive decrease in peak	 B1 B1 B1
5ai	Change in moment due to W = $mg \times (\Delta)L$	